

Lilly George

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EDUCATION:

Purdue University John Martinson Honors College: *West Lafayette, IN* **Class of 2028**
Dean's List/Semester Honors - Alpha Lambda Delta/Phi Eta Sigma Honors Society member **GPA 3.4**
Bachelor of Science, Mechanical Engineering, Minor Materials Science and Engineering

EXPERIENCE:

- Honda Motor Company – Mechanical Engineering Test Intern,** *Haw River, NC* **May 2026 – August 2026**
- Collected endurance test strain/acceleration data from lawnmower, calculated damage accrued in MATLAB, and correlated data to hydraulic actuator movement, condensing test time by half and increasing efficiency
 - Modeled (CATIA) and CAE-validated a lawnmower blade plate redesign to eliminate plate yielding during assembly, further validated through physical stress analysis; presented solution and resolved issue
 - Constructed drive system and circuit board for troubleshooting electric motor hall sensors and thermistors
 - Performed thermal study on prototype; ensured engine matching criteria met, compared to competitors
- Purdue Formula SAE Electric Racing – Vehicle Dynamics,** *Purdue University* **August 2024 – Present**
- Designed and set up Ansys FEA for 16 suspension tabs; cut tabs on rotary laser and created jig for welding
 - Analyzed carbon fiber suspension members in tension to extract stiffness/refine manufacturing for strength
 - Used MATLAB to optimize anti-roll bar geometry, material, and heat treatment based on stiffness targets
- Purdue Aerial Robotics Team – Testing Division Lead,** *Purdue University* **August 2026 - Present**
- Built thrust stand to validate propellor choice, measured servo slop to ensure stable control surfaces
 - Founded new club division from the group up, oversaw composite layup teaching and improvements
- Undergraduate Research Assistant,** *Purdue Layered Materials and Structures Lab* **January 2026 – Present**
- Initiated study of using new MAX precursors from other industries for mitigation of MXene purity issues
 - Investigated feasibility/properties of rare earth metal doping in MXenes for semiconductive applications
- Cummins – Mechanical Engineering Intern,** *Columbus, IN* **June 2025 – August 2025**
- Followed and supported a fast-paced 60L genset program; determined necessary engine validation tasks
 - Researched turbomachinery integration on larger alternative fuel (biodiesel, H₂) engines

LEADERSHIP:

- Mentor – Women in Engineering 1:1 Mentoring Program,** *Purdue University* **August 2025 – Present**
- Monthly meetings with freshman mentee to share advice and empower to foster success in college
- Volunteer - Pittsburgh Vintage Grand Prix,** *Pittsburgh, PA* **June 2015 – June 2024**
- Volunteered for three 16-hr days each summer at car shows; oversaw registration/parking of 300+ cars/day
- STUNT Team Member,** *Purdue University* **August 2024 – May 2025**
- Formulated inaugural team, attended two tournaments and contributed to first game victory
 - Practiced nine hours a week to learn 15 stunting and tumbling routines from August to February
- Moon Area Gymnastics Association – Coach,** *Moon Township, PA* **December 2019 - May 2024**
- Led 15-20 young gymnasts as part of a team to teach skill techniques and safety through patience, positive reinforcement, verbal corrections, and physical demonstrations to improve their gymnastics abilities
 - Choreographed/taught a 20 diverse gymnasts a 3-minute routine to highlight their skills in a showcase

SKILLS AND ACHIEVEMENTS:

Skills: Siemens NX/Teamcenter, MATLAB, UTM Testing, Manual Mill, Manual Lathe, Waterjet, Rotary Tube Laser, Strain Gauges, Accelerometers, Thermocouples, Ansys FEA, Autodesk Inventor, Fusion360, Catia V5, Python

Achievements: Commended by National Merit Scholarship Program (2024)